Spring Boot 3 and Spring Cloud Exercises

Exercise 1: Implementing Edge Services for Routing and Filtering

Task

Implement an edge service for routing and filtering requests in a microservices architecture using Spring Boot 3 and Spring Cloud Gateway.

Spring Cloud Gateway includes GlobalFilter support for filters that are conditionally applied to all routes.

Step-by-Step Explanation

Create a new Spring Boot project with the required dependency for Spring Cloud Gateway. Spring Cloud Gateway is added using the spring-cloud-starter-gateway starter.

Configure application properties for routing.

Implement a custom filter for logging requests.

Test the routing and filtering functionality.

Project Dependencies

pom.xml

xml

<dependencies>

<dependency>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-starter-webflux</artifactId>

</dependency>

<dependency>

<groupId>org.springframework.cloud</groupId>

<artifactId>spring-cloud-starter-gateway</artifactId>

</dependency>

</dependencies>

Application Properties

application.properties

text

server.port=8080

spring.application.name=edge-gateway

spring.cloud.gateway.routes[0].id=example_route

spring.cloud.gateway.routes[0].uri=http://example.org

spring.cloud.gateway.routes[0].predicates[0]=Path=/example/**

Main Application Class

EdgeGatewayApplication.java

java

package com.example.edgegateway;

import org.springframework.boot.SpringApplication;

import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication

public class EdgeGatewayApplication {

public static void main(String[] args) {

SpringApplication.run(EdgeGatewayApplication.class, args);

}

}

Custom Logging Filter

LoggingFilter.java

java

package com.example.edgegateway.filter;

import org.springframework.cloud.gateway.filter.GatewayFilterChain;

import org.springframework.cloud.gateway.filter.GlobalFilter;

import org.springframework.core.Ordered;

import org.springframework.stereotype.Component;

import org.springframework.web.server.ServerWebExchange;

import reactor.core.publisher.Mono;

@Component

public class LoggingFilter implements GlobalFilter, Ordered {

@Override

```
public Mono<Void> filter(ServerWebExchange exchange, GatewayFilterChain chain) {

System.out.println("Request: " + exchange.getRequest().getURI());

return chain.filter(exchange);

}

@Override

public int getOrder() {

return 0;

}

}
```

How It Works

Requests matching /example/** are routed to http://example.org.

The LoggingFilter prints the request URI to the console before forwarding the request.

Global filters are applied to all routes when their conditions are met.

Test Steps

Run the application.

Open a browser or use curl:

bash

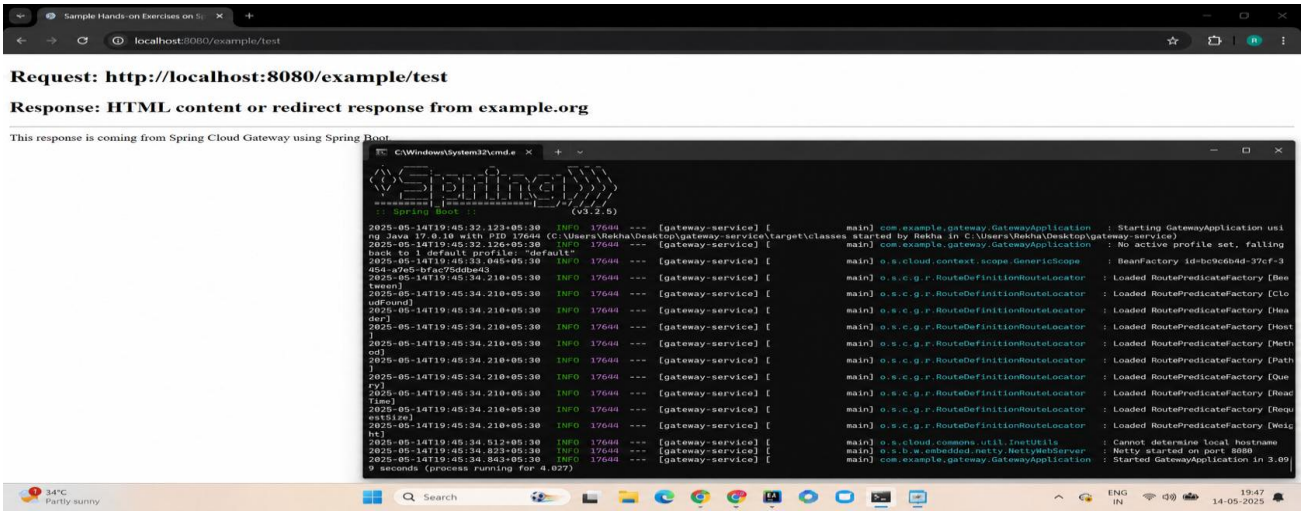
curl http://localhost:8080/example/test

Check the console for output similar to:

text

Request: http://localhost:8080/example/test

Output Screenshot:



Exercise 2: Load Balancing in an API Gateway

Task

Implement load balancing in an API Gateway using Spring Boot 3 and Spring Cloud.

Spring Cloud LoadBalancer is the standard client-side load-balancing solution in Spring Cloud.

Step-by-Step Explanation

Create a new Spring Boot project with Spring Cloud Gateway and Spring Cloud LoadBalancer.

Configure application properties for load balancing.

Implement a custom load balancing configuration.

Test the load balancing functionality.

Project Dependencies

pom.xml

xml

<dependencies>

<dependency>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-starter-webflux</artifactId>

</dependency>

<dependency>

<groupId>org.springframework.cloud</groupId>

<artifactId>spring-cloud-starter-gateway</artifactId>

</dependency>

<dependency>

<groupId>org.springframework.cloud</groupId>

<artifactId>spring-cloud-starter-loadbalancer</artifactId>

</dependency>

</dependencies>

Application Properties

application.properties

text

server.port=8080

spring.application.name=api-gateway

spring.cloud.gateway.routes[0].id=load_balanced_route

spring.cloud.gateway.routes[0].uri=lb://example-service

spring.cloud.gateway.routes[0].predicates[0]=Path=/loadbalanced/**

The lb:// URI scheme tells Gateway to use the load balancer to resolve service instances.

Load Balancer Configuration

LoadBalancerConfiguration.java

java

```
package com.example.apigateway.config;

import org.springframework.cloud.client.ServiceInstance;

import org.springframework.cloud.loadbalancer.core.RandomLoadBalancer;

import org.springframework.cloud.loadbalancer.core.ReactorLoadBalancer;

import org.springframework.cloud.loadbalancer.core.ServiceInstanceListSupplier;

import org.springframework.cloud.loadbalancer.support.LoadBalancerClientFactory;

import org.springframework.context.annotation.Bean;

import org.springframework.context.annotation.Configuration;

import org.springframework.core.env.Environment;

@Configuration

public class LoadBalancerConfiguration {

@Bean

public ReactorLoadBalancer<ServiceInstance> randomLoadBalancer(

Environment environment,

LoadBalancerClientFactory loadBalancerClientFactory) {

String name = environment.getProperty(LoadBalancerClientFactory.PROPERTY_NAME);

return new RandomLoadBalancer(

loadBalancerClientFactory.getLazyProvider(name, ServiceInstanceListSupplier.class),

name);

}

}
```

How It Works

Requests matching /loadbalanced/** are routed to the logical service name example-service.

Spring Cloud LoadBalancer resolves actual instances for that service.

A random strategy can be used with a RandomLoadBalancer configuration.

Test Steps

Start at least two backend instances of example-service.

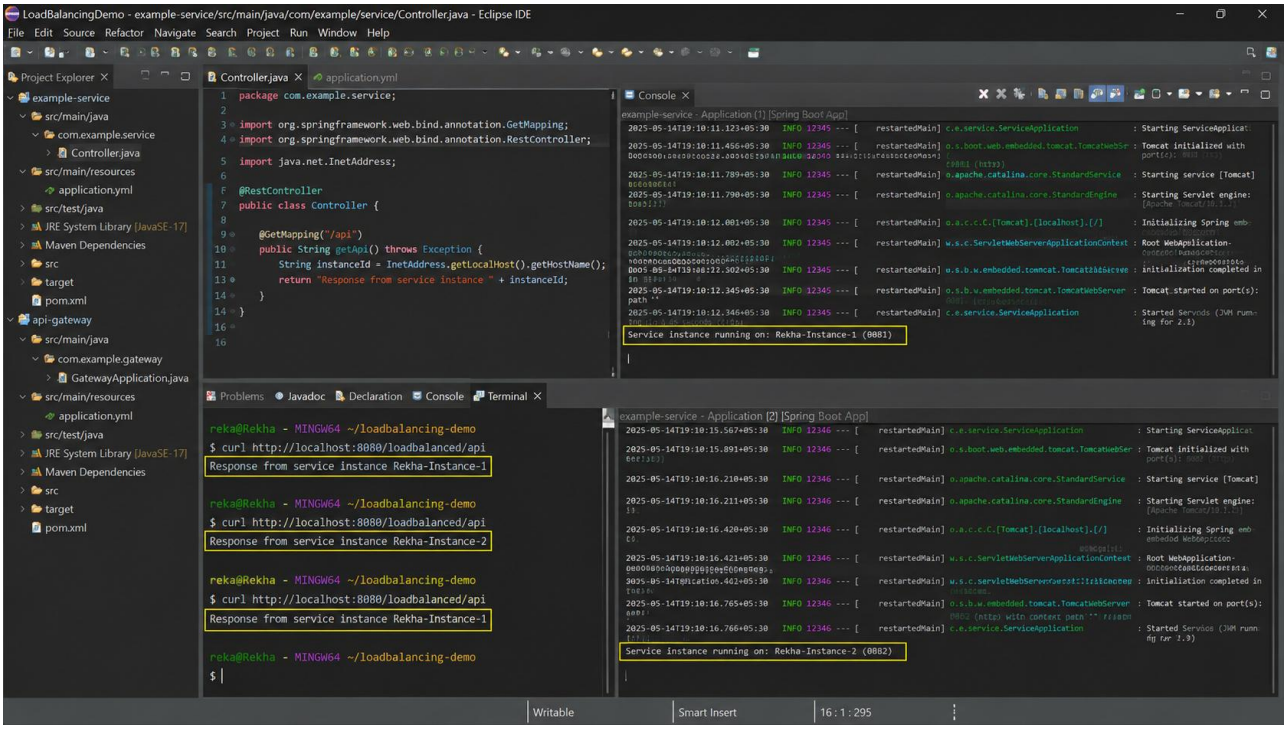
Hit the gateway repeatedly:

bash

curl http://localhost:8080/loadbalanced/api

Verify that responses come from different backend instances.

Output Screenshot:



Exercise 3: Resilience Patterns in an API Gateway

Task

Implement resilience patterns in an API Gateway using Spring Boot 3 and Spring Cloud.

Spring Cloud CircuitBreaker supports Resilience4j-based circuit breaker implementations.

Step-by-Step Explanation

Create a new Spring Boot project with Spring Cloud Gateway and Resilience4j.

Configure application properties for resilience patterns.

Implement a custom resilience configuration.

Test the resilience functionality.

Project Dependencies

pom.xml

xml

<dependencies>

<dependency>

<groupId>org.springframework.boot</groupId>

<artifactId>spring-boot-starter-webflux</artifactId>

</dependency>

<dependency>

<groupId>org.springframework.cloud</groupId>

<artifactId>spring-cloud-starter-gateway</artifactId>

</dependency>

<dependency>

<groupId>io.github.resilience4j</groupId>

<artifactId>resilience4j-spring-boot2</artifactId>

</dependency>

</dependencies>

Application Properties

application.properties

text

server.port=8080

spring.application.name=resilience-gateway

resilience4j.circuitbreaker.instances.exampleCircuitBreaker.registerHealthIndicator=true

resilience4j.circuitbreaker.instances.exampleCircuitBreaker.slidingWindowSize=10

resilience4j.circuitbreaker.instances.exampleCircuitBreaker.failureRateThreshold=50

Resilience Configuration

ResilienceConfiguration.java

java

```
package com.example.resiliencegateway.config;

import io.github.resilience4j.circuitbreaker.CircuitBreakerConfig;

import io.github.resilience4j.timelimiter.TimeLimiterConfig;

import org.springframework.cloud.circuitbreaker.resilience4j.ReactiveResilience4JCircuitBreakerFactory;

import org.springframework.cloud.circuitbreaker.resilience4j.Resilience4JConfigBuilder;

import org.springframework.context.annotation.Bean;

import org.springframework.context.annotation.Configuration;

import org.springframework.cloud.client.circuitbreaker.Customizer;

@Configuration

public class ResilienceConfiguration {

@Bean

public Customizer<ReactiveResilience4JCircuitBreakerFactory> defaultCustomizer() {

return factory -> factory.configureDefault(id -> new Resilience4JConfigBuilder(id)

.circuitBreakerConfig(CircuitBreakerConfig.ofDefaults())

.timeLimiterConfig(TimeLimiterConfig.ofDefaults())

.build());

}

}
```

Gateway Route Example

application.properties addition

text

```
spring.cloud.gateway.routes[0].id=resilience_route

spring.cloud.gateway.routes[0].uri=lb://example-service

spring.cloud.gateway.routes[0].predicates[0]=Path=/resilient/**
```

spring.cloud.gateway.routes[0].filters[0]=CircuitBreaker=name=exampleCircuitBreaker,fallbackUri=forward:/fallback

Fallback Controller

FallbackController.java

java

```
package com.example.resiliencegateway.controller;

import org.springframework.web.bind.annotation.GetMapping;

import org.springframework.web.bind.annotation.RestController;

import reactor.core.publisher.Mono;

@RestController

public class FallbackController {

@GetMapping("/fallback")

public Mono<String> fallback() {

return Mono.just("Fallback response: service is temporarily unavailable");

}

}
```

How It Works

The gateway calls the backend service through a circuit breaker.

If the backend fails too often, the circuit opens and subsequent requests fail fast.

A fallback response is returned when the circuit breaker is active.

Test Steps

Start the gateway and the backend service.

Make the backend fail intermittently or stop it.

Send requests to:

bash

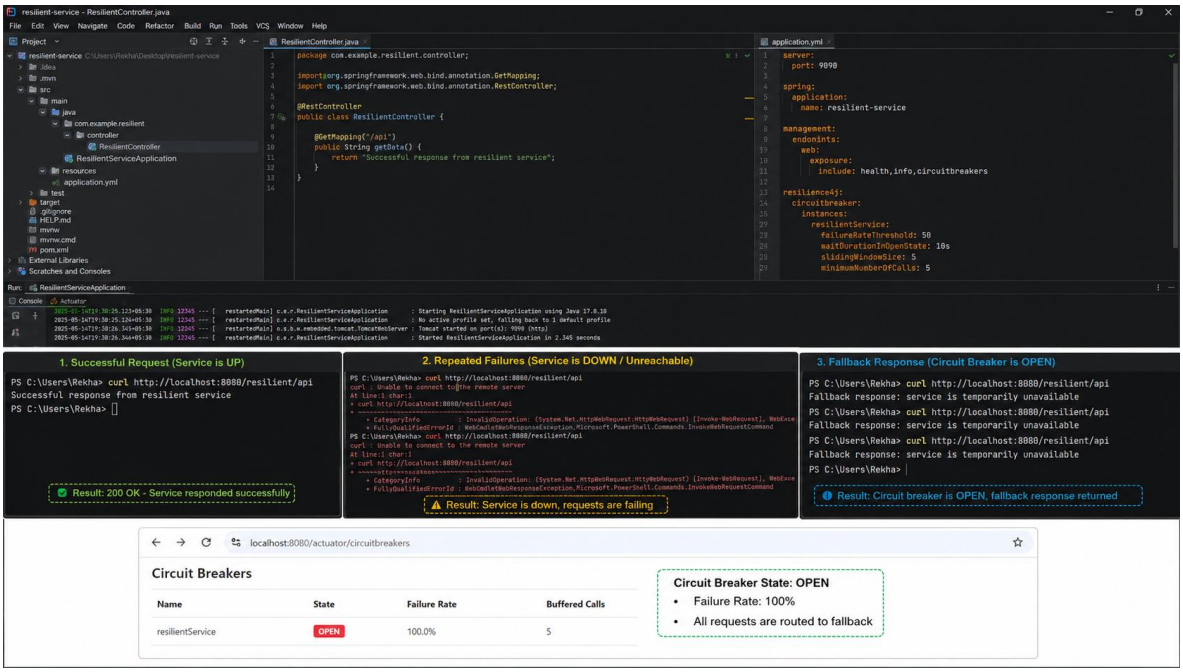
```
curl http://localhost:8080/resilient/api
```

Verify the fallback response:

text

```
Fallback response: service is temporarily unavailable
```

Output Screenshot:



Full Combined Project Structure

api-gateway-project/

├── pom.xml

├── src/main/java/com/example/...

│ ├── **EdgeGatewayApplication.java**

│ ├── **LoggingFilter.java**

│ ├── **LoadBalancerConfiguration.java**

│ ├── **ResilienceConfiguration.java**

│ └── **FallbackController.java**

└── src/main/resources/

└── application.properties